

10.2 Solution: Since the EM radiation with circular polarization but different helicity is scattered independently, the scattered EM radiation for elliptic polarization will be linear superposition of the scattered, circularly polarized radiation. In the long-wavelength limit, Eq. (10.71) becomes

$$\frac{d\sigma_{\text{sc}}}{d\Omega} = \frac{2\pi}{3} \frac{k^4 a^6}{1+r^2} \left| \left(\mathbf{X}_{1,1} - 2i\mathbf{n} \times \mathbf{X}_{1,1} \right) + re^{i\alpha} \left(\mathbf{X}_{1,-1} + 2i\mathbf{n} \times \mathbf{X}_{1,-1} \right) \right|^2.$$

By expanding the complex square, we have

$$\begin{aligned} & \left| \left(\mathbf{X}_{1,1} - 2i\mathbf{n} \times \mathbf{X}_{1,1} \right) + re^{i\alpha} \left(\mathbf{X}_{1,-1} + 2i\mathbf{n} \times \mathbf{X}_{1,-1} \right) \right|^2 \\ = & |\mathbf{X}_{1,1} - 2i\mathbf{n} \times \mathbf{X}_{1,1}|^2 + r^2 |\mathbf{X}_{1,-1} + 2i\mathbf{n} \times \mathbf{X}_{1,-1}|^2 \\ & + 2\text{Re} \left[\left(\mathbf{X}_{1,1} - 2i\mathbf{n} \times \mathbf{X}_{1,1} \right) \cdot re^{-i\alpha} \left(\mathbf{X}_{1,-1} + 2i\mathbf{n} \times \mathbf{X}_{1,-1} \right)^* \right]. \end{aligned}$$

Using the results between Eqs. (10.71) and (10.72), we know

$$|\mathbf{X}_{1,1} - 2i\mathbf{n} \times \mathbf{X}_{1,1}|^2 = |\mathbf{X}_{1,-1} + 2i\mathbf{n} \times \mathbf{X}_{1,-1}|^2 = \frac{15}{16\pi} (1 + \cos^2 \theta) - \frac{3}{2\pi} \cos \theta.$$

All we need to do now is to calculate the cross product term. To this end, we will need the explicit expression for $\mathbf{X}_{1,\pm 1}$.

The defition of $\mathbf{X}_{lm}$ is

$$\mathbf{X}_{lm} = \frac{1}{\sqrt{l(l+1)}} \mathbf{L} Y_{lm} = \frac{1}{\sqrt{l(l+1)}} (L_x Y_{lm}, L_y Y_{lm}, L_z Y_{lm})^\top.$$

Since $L_\pm = L_x \pm iL_y$, we know

$$L_x = \frac{1}{2}(L_+ + L_-), \quad L_y = \frac{1}{2i}(L_+ - L_-).$$

Also,

$$L_\pm Y_{lm} = \sqrt{(l \mp m)(l \pm m + 1)} Y_{l, m \pm 1},$$

then

$$\mathbf{X}_{lm} = \frac{1}{\sqrt{l(l+1)}} \begin{pmatrix} \frac{1}{2} \left(\sqrt{(l-m)(l+m+1)} Y_{l, m+1} + \sqrt{(l+m)(l-m+1)} Y_{l, m-1} \right) \\ \frac{1}{2i} \left(\sqrt{(l-m)(l+m+1)} Y_{l, m+1} - \sqrt{(l+m)(l-m+1)} Y_{l, m-1} \right) \\ m Y_{lm} \end{pmatrix}.$$

For $l = 1$ and $m = 1$, we have

$$\mathbf{X}_{1,1} = \left(\frac{1}{2} Y_{1,0}, \frac{i}{2} Y_{1,0}, \frac{1}{\sqrt{2}} Y_{1,1} \right)^\top,$$

and $l = 1$ and $m = -1$,

$$\mathbf{X}_{1,-1} = \left(\frac{1}{2} Y_{1,0}, -\frac{i}{2} Y_{1,0}, -\frac{1}{\sqrt{2}} Y_{1,-1} \right)^\top.$$

From the identity $Y_{lm}^* = (-1)^m Y_{l, -m}$, we know $Y_{1,1}^* = -Y_{1,-1}$, and $\mathbf{X}_{1,1}^* = \mathbf{X}_{1,-1}$. Here,

$$Y_{1,0} = \sqrt{\frac{3}{4\pi}} \cos \theta, \quad Y_{1,1} = -\sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi}.$$

Now, we can calculate the cross product term as

$$\begin{aligned}
& \left(\mathbf{X}_{1,1} - 2i\mathbf{n} \times \mathbf{X}_{1,1} \right) \cdot \left(\mathbf{X}_{1,-1} + 2i\mathbf{n} \times \mathbf{X}_{1,-1} \right)^* \\
&= \left(\mathbf{X}_{1,1} - 2i\mathbf{n} \times \mathbf{X}_{1,1} \right) \cdot \left(\mathbf{X}_{1,1} - 2i\mathbf{n} \times \mathbf{X}_{1,1} \right) \\
&= \mathbf{X}_{1,1} \cdot \mathbf{X}_{1,1} - 4i\mathbf{X}_{1,1} \cdot (\mathbf{n} \times \mathbf{X}_{1,1}) - 4(\mathbf{n} \times \mathbf{X}_{1,1}) \cdot (\mathbf{n} \times \mathbf{X}_{1,1}) \\
&= -3\mathbf{X}_{1,1} \cdot \mathbf{X}_{1,1},
\end{aligned}$$

where we have used the fact that $\mathbf{X}_{1,1} \cdot (\mathbf{n} \times \mathbf{X}_{1,1}) = 0$ and $(\mathbf{n} \times \mathbf{X}_{1,1}) \cdot (\mathbf{n} \times \mathbf{X}_{1,1}) = \mathbf{X}_{1,1} \cdot \mathbf{X}_{1,1}$. Finally,

$$\mathbf{X}_{1,1} \cdot \mathbf{X}_{1,1} = \frac{1}{2}Y_{1,1}^2 = \frac{3}{16\pi} \sin^2 \theta e^{2i\phi},$$

and

$$\begin{aligned}
& 2\text{Re} \left[\left(\mathbf{X}_{1,1} - 2i\mathbf{n} \times \mathbf{X}_{1,1} \right) \cdot r e^{-i\alpha} \left(\mathbf{X}_{1,-1} + 2i\mathbf{n} \times \mathbf{X}_{1,-1} \right)^* \right] \\
&= 2\text{Re} \left[r e^{-i\alpha} \cdot \frac{-9}{16\pi} \sin^2 \theta e^{2i\phi} \right] \\
&= -\frac{9}{8\pi} r \sin^2 \theta \cos(2\phi - \alpha).
\end{aligned}$$

Putting everything together, we will have

$$\begin{aligned}
\frac{d\sigma_{\text{sc}}}{d\Omega} &= \frac{2\pi}{3} \frac{k^4 a^6}{1+r^2} \left[\left(\frac{15}{16\pi} (1 + \cos^2 \theta) - \frac{3}{2\pi} \cos \theta \right) + r^2 \left(\frac{15}{16\pi} (1 + \cos^2 \theta) - \frac{3}{2\pi} \cos \theta \right) \right. \\
&\quad \left. - \frac{9}{8\pi} r \sin^2 \theta \cos(2\phi - \alpha) \right] \\
&= \frac{2\pi}{3} k^4 a^6 \left[\frac{15}{16\pi} (1 + \cos^2 \theta) - \frac{3}{2\pi} \cos \theta - \frac{9}{8\pi} \left(\frac{r}{1+r^2} \right) \sin^2 \theta \cos(2\phi - \alpha) \right] \\
&= k^4 a^6 \left[\frac{5}{8} (1 + \cos^2 \theta) - \cos \theta - \frac{3}{4} \left(\frac{r}{1+r^2} \right) \sin^2 \theta \cos(2\phi - \alpha) \right].
\end{aligned}$$

The same expression can also be obtained by applying the result from Prob. 10.1 (a). With $\mathbf{n}_0 = (0, 0, 1)$, $\mathbf{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, $\boldsymbol{\epsilon}_1 = (1, 0, 0)$, $\boldsymbol{\epsilon}_2 = (0, 1, 0)$, $\boldsymbol{\epsilon}_{\pm} = (\boldsymbol{\epsilon}_1 \pm i\boldsymbol{\epsilon}_2)/\sqrt{2} = (1, \pm i, 0)/\sqrt{2}$, and

$$\boldsymbol{\epsilon} = \frac{1}{\sqrt{1+r^2}} (\boldsymbol{\epsilon}_+ + r e^{i\alpha} \boldsymbol{\epsilon}_-) = \frac{1}{\sqrt{2(1+r^2)}} (1 + r e^{i\alpha}, i(1 - r e^{i\alpha}), 0),$$

we can perform the vector products as

$$\mathbf{n} \cdot \boldsymbol{\epsilon} = \frac{1}{\sqrt{2(1+r^2)}} ((1 + r e^{i\alpha}) \sin \theta \cos \phi + i(1 - r e^{i\alpha}) \sin \theta \sin \phi),$$

$$|\mathbf{n} \cdot \boldsymbol{\epsilon}|^2 = \frac{1}{2(1+r^2)} \left((1 + 2r \cos \alpha + r^2) \sin^2 \theta \cos^2 \phi + (1 - 2r \cos \alpha + r^2) \sin^2 \theta \sin^2 \phi \right)$$

$$\begin{aligned}
& +4r \sin^2 \theta \sin \phi \cos \phi \sin \alpha \Big) \\
& = \frac{1}{2(1+r^2)} \left((1+r^2) \sin^2 \theta \cos^2 \phi + 2r \sin^2 \theta \cos(2\phi) \cos \alpha + 2r \sin^2 \theta \sin(2\phi) \sin \alpha \right) \\
& = \frac{1}{2} \sin^2 \theta + \frac{r}{1+r^2} \sin^2 \theta \cos(2\phi - \alpha),
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{n}_0 \times \boldsymbol{\epsilon} &= \frac{1}{\sqrt{2(1+r^2)}} \left(-i(1 - re^{i\alpha}), (1 + re^{i\alpha}), 0 \right), \\
|\mathbf{n} \cdot (\mathbf{n}_0 \times \boldsymbol{\epsilon})|^2 &= \frac{1}{2(1+r^2)} \left((1 - 2r \cos \alpha + r^2) \sin^2 \theta \cos^2 \phi + (1 + 2r \cos \alpha + r^2) \sin^2 \theta \sin^2 \phi \right. \\
&\quad \left. - 4r \sin^2 \theta \sin \phi \cos \phi \sin \alpha \right) \\
&= \frac{1}{2(1+r^2)} \left((1+r^2) \sin^2 \theta - 2r \sin^2 \theta \cos(2\phi) \cos \alpha - 2r \sin^2 \theta \sin(2\phi) \sin \alpha \right) \\
&= \frac{1}{2} \sin^2 \theta - \frac{r}{1+r^2} \sin^2 \theta \cos(2\phi - \alpha).
\end{aligned}$$

Then, using Prob. 10.1 (a),

$$\begin{aligned}
\frac{d\sigma_{\text{sc}}}{d\Omega} &= k^4 a^6 \left[\frac{5}{4} - |\mathbf{n} \cdot \boldsymbol{\epsilon}|^2 - \frac{1}{4} |\mathbf{n} \cdot (\mathbf{n}_0 \times \boldsymbol{\epsilon})|^2 - \mathbf{n} \cdot \mathbf{n}_0 \right] \\
&= k^4 a^6 \left[\frac{5}{4} - \frac{1}{2} \sin^2 \theta - \frac{r}{1+r^2} \sin^2 \theta \cos(2\phi - \alpha) \right. \\
&\quad \left. - \frac{1}{4} \left(\frac{1}{2} \sin^2 \theta - \frac{r}{1+r^2} \sin^2 \theta \cos(2\phi - \alpha) \right) - \cos \theta \right] \\
&= k^4 a^6 \left[\frac{5}{4} - \frac{5}{8} \sin^2 \theta - \cos \theta - \frac{3}{4} \left(\frac{r}{1+r^2} \right) \sin^2 \theta \cos(2\phi - \alpha) \right] \\
&= k^4 a^6 \left[\frac{5}{8} (1 + \cos^2 \theta) - \cos \theta - \frac{3}{4} \left(\frac{r}{1+r^2} \right) \sin^2 \theta \cos(2\phi - \alpha) \right].
\end{aligned}$$